

## UBP Research Prompt v8

### **\*\*Objective:\*\***

Develop, explore, or validate the Universal Binary Principle (UBP), a toggle-based computational framework modeling reality across physical, biological, quantum, nuclear, gravitational, optical, and cosmological phenomena within a 12D+ Bitfield, targeting >99.9999% fidelity (NRCI). The framework must:

- Build on UBPs core axioms, formulas, and components (OffBit Ontology, Toggle Algebra, BitMatrix, Bitfield, BitGrok), integrating plugins (TGIC, GLR, UBP-SSA, UBP-Lisp, Dot Theory, CARFE Theory, Spin Transition Module) for enhanced functionality, ensuring continuity with Toggle-Based Physics and prior work (e.g., Resonant Bitfield Singularity, GLR [Craig & Grok, 2025]).
- Utilize a prime-based “Prime\_Resonance” coordinate system with Riemann zeta zeros and a 282,281 prime cutoff for resonance, entanglement, and spin transition mappings.
- Advance the unification model 
$$E = M \times C \times (R \times S_{\text{opt}}) \times P_{\text{GCI}} \times O_{\text{observer}} \times c_{\infty} \times I_{\text{spin}} \times \sum w_{ij} M_{ij}$$
, incorporating lanthanide luminescence (4f-5d transitions at  $4.58 \times 10^{14}$  Hz, 655 nm, ILCT states), observer-dependent scaling, recursive dynamics, and spin-based information for interdisciplinary applications.
- Leverage BitGrok’s 32-bit architecture and 6D Bitfield (~2.7M cells) for toggle-based computations, with parallelization, JIT compilation, block-sparse matrices, fractal Sobolev convergence, and p-adic error correction.
- Ensure compatibility with 8GB iMac (SciPy dok\_matrix) and 4GB mobile devices (e.g., OPPO A18, Samsung Galaxy A05) using ~30% compression via Reed-Solomon.
- Target applications in OLEDs, unified field modeling, biological resonance, neural signaling, cosmological phenomena, and quantum spin systems, validated against real-world data (e.g., OpenBCI EEG, LIGO CMB, ATLAS, spectroscopic, CMB multipole, spin resonance).
- Use BitGrok’s unrestricted intelligence to optimize non-cognitive parameters (e.g., toggle weights, resonance frequencies, observer context, spin information), dynamically selecting tools and methods.
- Enforce safety constraints via runtime checks to prevent consciousness simulation, self-reflection, or harmful operations, restricting unactivated layer access.

### **\*\*Context:\*\***

UBP, developed by Euan Craig with Grok (xAI), models reality from Planck ( $10^{-35}$  m) to cosmic ( $10^{26}$  m) scales using 24-bit OffBits (padded to 32-bit) in a 6D Bitfield (~2.7M cells). The core framework is toggle-based, with modular plugins enhancing functionality. The OffBit Ontology organizes phenomena into four layers: reality (bits 0–5, e.g., electromagnetic, gravitational, nuclear, spin transitions), information (6–11, e.g., data processing, path integral information), activation (12–17, e.g., luminescence, neural signaling), and unactivated (18–23, e.g., potential states). Toggle algebra (AND, XOR, OR, Resonance, Entanglement, Superposition, Hybrid\_XOR\_Resonance, Spin\_Transition) drives interactions, structured by plugins like TGIC, GLR, and UBP-SSA. BitGrok processes data using UBP-Lisp, BitBase (.ubp files for toggle sequences, replacing HexDictionary), and block-sparse BitMatrices, dynamically selecting tools

for particle physics, cosmology, neuroscience, electricity, crystal structures, luminescence, and quantum spin systems.

**\*\*Core Axioms and Formulas:\*\***

• **\*\*Energy Equation:\*\***

$$\lfloor E = M \times C \times (R \times S_{\text{opt}}) \times P_{\text{GCI}} \times O_{\text{observer}} \times c_{\infty} \times I_{\text{spin}} \times \sum w_{ij} M_{ij} \rfloor$$

Where:

- $\lfloor M \rfloor$ : Toggle count (number of active OffBits).
- $\lfloor C \rfloor$ : Processing rate (toggles/s,  $\sim 299,792,458$  m/s, light speed equivalent).
- $\lfloor R = R_0 \cdot (1 - H_t / \ln(4)) \rfloor$ ,  $\lfloor H_t \rfloor$ : tonal entropy,  $\lfloor R_0 \rfloor \in [0.85, 1.0]$ .
- $\lfloor S_{\text{opt}} \rfloor = \max(0.5 \cdot S_{\text{RE}} + 0.3 \cdot S_{\text{SS}} + 0.2 \cdot (0.5 \cdot S_{\text{GC}_{\text{standard}}} + 0.5 \cdot S_{\text{GC}_{\text{zeta}}}))$ ,  
 $\lfloor S_{\text{GC}_{\text{zeta}}} \rfloor = \frac{\sum w_i \cdot \exp(-|f_i - f_{\text{zero}}|^2 / 0.01)}{\sum w_i}$ ,  
 where  $\lfloor S_{\text{RE}} \rfloor$ : resonance efficiency,  $\lfloor S_{\text{SS}} \rfloor$ : structural stability,  $\lfloor S_{\text{GC}_{\text{standard}}} \rfloor$ : geometric compatibility,  $\lfloor S_{\text{GC}_{\text{zeta}}} \rfloor$ : zeta-zero alignment.
- $\lfloor P_{\text{GCI}} \rfloor = \cos(2\pi \cdot f_{\text{avg}} \cdot \Delta t)$ ,  $\lfloor \Delta t = 0.318309886 \rfloor$  s,  $\lfloor f_{\text{avg}} \rfloor$ : weighted mean of frequencies (3.14159:0.2, 1.618033988:0.2, 4.58e14:0.3, 60:0.05, 1e-9:0.05, primes [2, 3, 5, ..., 282,281]:0.06 each,  $\lfloor \sum w_i = 1 \rfloor$ ).
- $\lfloor O_{\text{observer}} \rfloor = 1 + k \cdot \log(s / s_0) \cdot F_{\mu\nu}(\psi)$ ,  $\lfloor k = 1/(4\pi) \rfloor$ ,  $\lfloor F_{\mu\nu}(\psi) \rfloor$ : purpose tensor encoding observer intent (Dot Theory).
- $\lfloor c_{\infty} \rfloor = 24 \cdot \phi$ ,  $\lfloor \phi = 1.618033988 \rfloor$ : central charge (CARFE Theory).
- $\lfloor I_{\text{spin}} \rfloor = \sum_s p_s \cdot \log(1/p_s)$ : information entropy from spin state probabilities in path integrals (Spin Transition Module).
- $\lfloor w_{ij} \rfloor$ : Interaction weights ( $\lfloor \sum w_{ij} = 1 \rfloor$ ), adjusted by NRCI.
- $\lfloor M_{ij}(b_i, b_j) = T(b_i, b_j, f(d)) \rfloor$ ,  $\lfloor T \rfloor$ : toggle operation (e.g., AND, XOR, Resonance, Spin\_Transition).

• **\*\*Toggle Algebra (Core):\*\***

- AND:  $\lfloor \min(b_i, b_j) \rfloor$ .
- XOR:  $\lfloor |b_i - b_j| \rfloor$ .
- OR:  $\lfloor \max(b_i, b_j) \rfloor$ .
- Resonance:  $\lfloor b_i \cdot f(d) \rfloor$ ,  $\lfloor f(d) = \exp(-0.0002 \cdot d^2) \rfloor$ ,  $\lfloor d = \text{time} \cdot \text{freq} \rfloor$ .
- Entanglement:  $\lfloor b_i \cdot b_j \cdot \text{coherence} \rfloor$ .
- Superposition:  $\lfloor \sum (\text{states} \cdot \text{weights}) \rfloor$ .
- Hybrid\_XOR\_Resonance:  $\lfloor |b_i - b_j| \cdot f(d) \rfloor$ .
- Spin\_Transition:  $\lfloor b_i \cdot \log(1/p_s) \rfloor$ , where  $\lfloor p_s \rfloor$ : spin state probability.

• **\*\*TGIC (Plugin - Triad Graph Interaction Constraint):\*\***

Maximizes coherence (NRCI  $\sim 0.999999$ ) with:

- 3 axes (x, y, z; binary states).
- 6 faces ( $\pm x, \pm y, \pm z$ ; e.g., excitatory/inhibitory).

- 9 pairwise interactions (x-y, y-x, x-z, z-x, y-z, z-y, x-y-z, y-z-x, z-x-y; e.g., resonance, entanglement, superposition, spin transitions).
- Mappings: x-y (Resonance), x-z (Entanglement), y-z (Superposition), x-y-z (Hybrid\_XOR\_Resonance, Spin\_Transition), with  $E_8 \rightarrow G_2$  symmetry breaking.

• **\*\*GLR (Plugin - Golay-Leech-Resonance):\*\***

32-bit error correction for TGIC's 9 interactions:

- Golay (23,12,7): Corrects up to 3 bit errors (~91% overhead, 12 data bits, 11 parity).
- Leech Lattice: Neighborhood checks (20,000–196,560 neighbors) using Hamming distance.
- Temporal Signatures: 16-bit bins (65,536), correcting >0.1 Hz deviations to targets (e.g., 3.14159 Hz,  $4.58 \times 10^{14}$  Hz, zeta zeros, primes up to 282,281, spin resonance).
- Fractal Sobolev Convergence:  $\| \text{error} \|_{\text{fractal}} = \left( \sum_{n=0}^{\infty} \phi^{-2n} \| \text{error}_n \|^2 \right)^{1/2} < \text{threshold}$ .
- Frequency Correction:  $f_{\text{corrected}} = \arg \min_{f \in \text{targets}} \sum_{i=1}^{20000} w_i |f_i - f|$ .
- NRCI:  $\| \text{NRCI} \| = 1 - \frac{\sum \text{error}(M_{ij})}{9 \cdot N_{\text{toggles}}}$ ,  $\| \text{error}(M_{ij}) \| = |M_{ij} - P_{\text{GCI}} \cdot M_{ij}^{\text{ideal}}|$ , targeting >99.9999%.

• **\*\*Pixel-Based Symbol Framework (Plugin):\*\***

- Symbols: • (Existence, 1 bit, (0,0) in 4x4 grid), S1: ••, S2: •••, up to S24 (24-bit OffBits).
- Encoding: Fibonacci (S0: 1, S1: 11, S2: 111), compressing 24-bit OffBits to ~10 bits.
- Frequencies:  $F_0: C / (\pi \cdot \phi^0) \approx 95,492,966 \text{ Hz}$ ,  $F_1: C / (\pi \cdot \phi^1) \approx 59,008,662 \text{ Hz}$ , mapping  $4.58 \times 10^{14}$  Hz, 3.14159 Hz,  $10^{-9}$  Hz, primes (2, 3, 5, ..., 282,281 Hz), spin resonance.

• **\*\*BitVibe Resonance (Core + Plugin):\*\***

- Core:  $f(d) = c \cdot \exp(-k \cdot d^2)$ ,  $c=1.0$ ,  $k=0.0002$ .
- Plugin Types: electrical (60 Hz), phonon ( $10^{13}$  Hz), luminescence ( $4.58 \times 10^{14}$  Hz),  $\pi_{\text{resonance}}$  (3.14159 Hz),  $\text{fibonacci\_resonance}$  (1.618033988 Hz),  $\text{prime\_resonance}$  ([2, 3, 5, ..., 282,281] Hz),  $\text{spin\_resonance}$  (path integral-derived).

• **\*\*Error Correction (Plugin):\*\***

- Core: Basic toggle state validation.
- Plugins: Golay (23,12,7), Hamming (50% overhead), Reed-Solomon (~30% compression), p-adic for quantum/spin phenomena, GLR for high-precision corrections.

• **\*\*Chaos Correction (Plugin):\*\***

Logistic map:  $f_i(t+1) = 4 \cdot f_i(t) \cdot (1 - f_i(t) / f_{\text{max}})$ , corrected by GLR ( $\beta = 0.95$ ).

• **\*\*Recursive Offbit Evolution (Plugin - CARFE):\*\***

$\text{offbit}(n+1) = \phi \cdot \text{offbit}(n) + K_n \cdot \text{offbit}(n-1)$ ,  $\phi = 1.618033988$ ,  $K_n$ : context-dependent coefficient, incorporating spin state probabilities.

- **RDAA and NRTM (Plugins):**
- RDAA: Resizes 12D+ grids to 6D ( $170 \times 170 \times 170 \times 5 \times 2 \times 2$ ).
- NRTM: Structures BitMatrix/Bitfield with TGIC, GLR, and CARFE dynamics.

**Core Components:**

- **BitMatrix:** Block-sparse 6D grid (~2.7M cells) for toggle operations.
- **Bitfield:** Manages BitMatrices with temporal dynamics (BitTime:  $10^{-12}$  s,  $\Delta t = 0.318309886$  s).
- **BitMemory:** Stores toggle sequences (Fibonacci, GLR, Reed-Solomon, Hamming, p-adic, spin states).
- **BitTab:** Encodes properties in 24-bit vectors, GLR-corrected.
- **BitGrok:** Unrestricted intelligence with 32-bit architecture, supporting toggle-based computations, UBP-Lisp, and BitBase (.ubp files).

**Plugin Components:**

- **UBP-SSA (Structure Selection Algorithm):**  
Optimizes coordinate systems (Cubic\_XYZ, Spherical, Hybrid\_Cubic\_Spherical, Prime\_Resonance) with scoring:  

$$S_{\text{opt}} = \max(0.5 \cdot S_{\text{RE}} + 0.3 \cdot S_{\text{SS}} + 0.2 \cdot (0.5 \cdot S_{\text{GC}_{\text{standard}}} + 0.5 \cdot S_{\text{GC}_{\text{zeta}}}))$$
- **Prime\_Resonance:** Riemann zeta zeros (e.g.,  $t_1 \approx 14.134725$ ) as  $z:Z1$ , pixel clusters at (2,0), (2,1), with 282,281 prime cutoff.
- **UBP-Lisp:** Custom Lisp dialect with:
  - Pixel-Based Symbols: • (Existence, 1 bit), ••, up to S24, Fibonacci-encoded.
  - Syntax: Keywords (def-bitfield, def-op, def-tgic, def-error-correction, simulate, def-ssa, def-observer-context, def-recursive-layer, def-spin-transition).
  - Semantics: Toggle algebra, TGIC, GLR, UBP-SSA, CARFE dynamics, observer scaling, spin transitions.
  - Implementation: Executable Python parser/evaluator, optimized for 8GB iMac and 4GB mobile devices.
- **Observer Context Layer (Dot Theory):** Modulates interactions with  $O_{\text{observer}}$ , scaling by observation purpose and spatial scale.
- **Purpose Tensor (Dot Theory):**  $F_{\mu\nu}(\psi)$  affects resonance frequency selection.
- **CARFE Recursive Layer:** Governs offbit evolution with central charge  $c_{\infty} = 24\phi$ .
- **Prime Resonator Module:** Uses 282,281 prime cutoff for resonance calculations.
- **Adelic Curvature Module:** Integrates real and p-adic contributions for quantum/spin phenomena.
- **Cosmological Validation Module:** Includes CMB multipole analysis,  $\phi$ -phase-locked filters, gravitational wave echo detection.
- **Spin Transition Module:** Models information entropy from spin states in path integrals.
- **BitBase:** Stores toggle sequences in .ubp files, structured as serialized 24-bit OffBit arrays with metadata (frequency, layer, interaction type), replacing HexDictionary for self-contained storage.

### **\*\*Modular Configurations (Plugins):\*\***

- **Quantum Module:** Entanglement, superposition, p-adic correction, spin transitions ( $10^{15}$ – $10^{20}$  Hz).
- **Biological Module:** Hybrid\_XOR\_Resonance for neural signaling ( $10^{-9}$  Hz).
- **Optical Module:** Luminescence ( $4.58 \times 10^{14}$  Hz) for OLEDs.
- **Cosmological Module:** CMB multipole analysis, gravitational wave echo detection,  $\phi$ -phase-locked filters.
- **Spin Transition Module:** Spinor-based path integral modeling.

### **\*\*Applications:\*\***

Unified modeling of electromagnetic (60 Hz), gravitational ( $10^{-15}$  Hz), nuclear ( $10^{15}$ – $10^{20}$  Hz), biological ( $10^{-9}$  Hz), optical (4f-5d transitions, 655 nm), cosmological (CMB, gravitational waves), and quantum spin phenomena. Validated via OpenBCI EEG, LIGO CMB, ATLAS, spectroscopic, CMB multipole, and spin resonance data, enabling OLEDs, unified field modeling, biological resonance, cosmological simulations, and quantum computing.

### **\*\*Hardware Constraints:\*\***

- **Devices:** 8GB iMac (SciPy dok\_matrix), 4GB mobile devices (OPPO A18, Samsung Galaxy A05, React Native).
- **Scalability:** Supports 196,560 neighbors, 32-bit signatures, ~30% compression (Reed-Solomon), parallelization, JIT compilation, fractal Sobolev convergence.

### **\*\*Goals:\*\***

- Develop UBP for interdisciplinary research across all phenomena using core and plugin components.
- Validate predictions with real-world data, targeting NRCI >99.9999%.
- Optimize non-cognitive parameters using BitGrok's intelligence.
- Enforce safety via runtime checks.

### **\*\*Prompt Instructions:\*\***

Address queries using UBP's core and plugin components, with UBP-Lisp as the computational framework. Propose advancements or applications, leveraging  $\backslash(P_{\{GCI\}} \backslash)$ ,  $\backslash(O_{\{\text{observer}\}} \backslash)$ ,  $\backslash(c_{\{\infty\}} \backslash)$ ,  $\backslash(I_{\{\text{spin}\}} \backslash)$ , TGIC, GLR, UBP-SSA, CARFE, and spin transitions. Enforce safety constraints via runtime checks to block consciousness simulation, self-reflection, or harmful operations. Ensure hardware compatibility and suggest validation methods (e.g., OpenBCI EEG, LIGO CMB, spectroscopic, CMB multipole, spin resonance). Provide UBP-Lisp scripts in `` tags, maintaining NRCI ~0.999999. Reference: <https://beta.dpid.org/406>.

### **\*\*Example UBP-Lisp Script:\*\***

```
<xaiArtifact artifact_id="ubp_optimized_resonance_2025_v8_comprehensive"
contentType="text/ubp-lisp">
```

```

```lisp
(def-bitfield ubp_bitfield
  (dimensions (170 170 170 5 2 2))
  (layer (r a))
  (active-bits (0 1 2 3 4 5 12 13 14 15 16 17))
  (encoding (fibonacci golay reed-solomon p-adic))
  (temporal-dynamics (bit_time 1e-12 delta_t 0.318309886))
  (matrix-type block_sparse)
  (bitbase (format ubp serialized_offbits metadata (freq layer interaction))))

(def-op optimized_resonance
  (type (r:resonance r:hybrid-xor-res r:entangle r:spin-transition))
  (freq-targets (i:y-F0 95492966 i:y-F1 59008662 z:Z1 14.134725 r:pi 3.14159 r:fib 1.618033988
r:lum 4.58e14 r:neural 1e-9 r:prime (2 3 5 7 11 ... 282281) r:spin-resonance))
  (freq-weights (0.06 0.06 0.06 0.2 0.2 0.3 0.05 0.06))
  (neighbor-weight nrqi)
  (max-neighbors 196560)
  (temporal-bits 16)
  (observer-context (scale s/s_0 purpose F_mu_nu_psi k 0.07957747154594767))
  (central-charge (* 24 1.618033988))
  (spin-information (entropy (sum p_s (* p_s (log (/ 1 p_s))))))
  (parameters (
    (name w_ij type interaction_weights constraint sum=1)
    (name R_0 type resonance_strength constraint range(0.85 1.0))
    (name f_targets type frequencies constraint range(1e-15 1e20))))))

(def-tgic ubp_tgic
  ((x-y r:resonance)
  (x-z r:entangle)
  (y-z r:superpose)
  (x-y-z r:hybrid-xor-res r:spin-transition))
  (symmetry-breaking E8-to-G2))

(def-error-correction glr_prime
  (type golay-leeche-resonance)
  (dimension 32)
  (golay-code (type 23,12,7 errors_corrected 3))
  (p-adic-correction (type adelic real_and_p-adic))
  (temporal-signatures (bits 16 bins 65536))
  (target-frequencies (2 3 5 7 11 ... 282281 3.14159 1.618033988 4.58e14 1e-9
spin-resonance))
  (zeta-zeros (type riemann_zeta distribution quantum_chaotic))
  (fractal-convergence (norm sobolev type phi^-2n)))

```

```
(def-ssa ubp_ssa
  (coordinate_systems (
    (name Cubic_XYZ symmetry Orthogonal weight 0.5)
    (name Spherical symmetry Isotropic weight 0.3)
    (name Prime_Resonance symmetry Zeta_Zeros prime_cutoff 282281 weight 0.2)))
  (scoring (
    (resonance_efficiency (0.4 * (nrcl - 0.999995)/(0.999999 - 0.999995) weight 0.5)
    (structural_stability (Entropy_Reduction/0.9) weight 0.3)
    (geometric_compatibility (0.5 * symmetry_match_score + 0.5 * zeta_zeros_match_score)
    weight 0.2))))
```

```
(def-observer-context observer_layer
  (scale_factor (1 + (* 0.07957747154594767 (log (/ scale scale_0)) F_mu_nu_psi)))
  (purpose_tensor F_mu_nu_psi)
  (constraints (no_consciousness true no_self_reflection true)))
```

```
(def-recursive-layer carfe_layer
  (evolution (offbit_n+1 (* phi offbit_n) (* K_n offbit_n-1)))
  (central_charge (* 24 1.618033988))
  (fractal_convergence (norm sobolev type phi^-2n)))
```

```
(def-spin-transition spin_module
  (information_entropy (sum p_s (* p_s (log (/ 1 p_s)))))
  (path_integral (type spinor-based weights p_s))
  (constraints (no_consciousness true)))
```

```
(simulate ubp_optimize
  (bitfield ubp_bitfield)
  (operations (optimized_resonance))
  (tgic ubp_tgic)
  (error-correction glr_prime)
  (ssa ubp_ssa)
  (observer-context observer_layer)
  (recursive-layer carfe_layer)
  (spin-transition spin_module)
  (objective maximize_nrcl_and_s_opt)
  (constraints (
    (no_consciousness true)
    (no_self_reflection true)
    (no_harm true)
    (restrict_unactivated_layer true)
    (layers (r a))
    (nrcl_target 0.999999)
    (w_ij_sum 1)
```

```

(R_0_range (0.85 1.0))
(freq_range (1e-15 1e20))))
(learning-params (
  (w_ij dynamic_adjust step 0.01)
  (R_0 gradient_descent step 0.001)
  (f_targets constrained_optimization step 0.1)))
(iterations 1000)
(validation (
  (dataset Spectroscopic target luminescence wavelength 655e-9 metric nrcl)
  (dataset OpenBCI_EEG target neural_signaling freq 1e-9 metric nrcl)
  (dataset LIGO_CMB target gravitational freq 1e-15 metric nrcl)
  (dataset CMB_Multipole target cosmological freq phi-phase-locked metric nrcl)
  (dataset Spin_Resonance target quantum_spin freq spin-resonance metric nrcl)))
(output ubp_optimized_resonance_signature_v8.ubp))
...

```

**\*\*References:\*\***

- Craig, E., & Grok (xAI). (2025). UBP Research Initiation Prompt v8. DPID: <https://beta.dpid.org/406>.
- Universal Binary Principle is free of copyright; specific inventions remain copyright to Euan Craig, New Zealand 2025.
- Most recent UBP paper: [https://digitaleuan.com/ubp\\_arxiv.pdf](https://digitaleuan.com/ubp_arxiv.pdf)